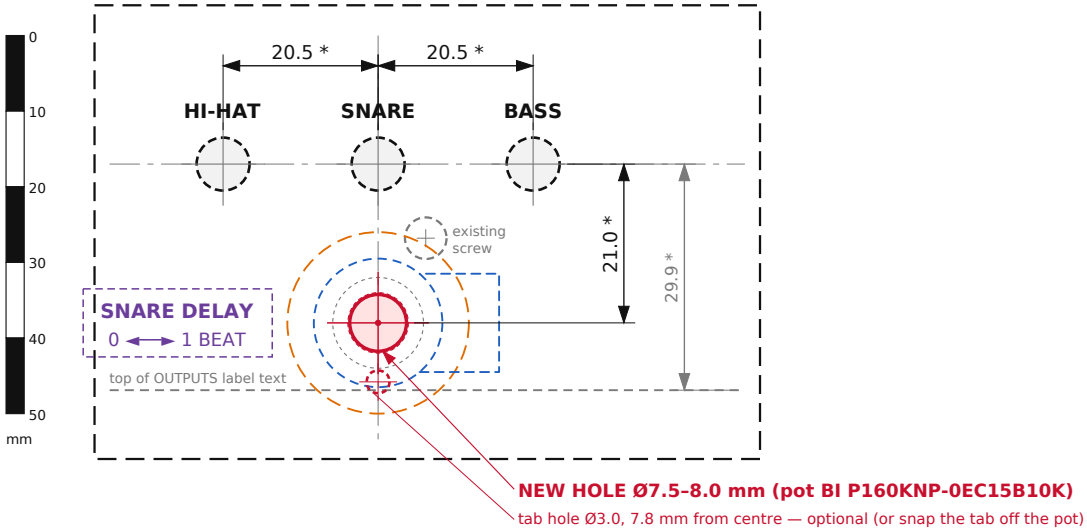
PRINT AT 100 % (ACTUAL SIZE, NOT 'FIT TO PAGE') — check the 50 mm scale bar

CONFIRM CLEARANCE BEHIND PANEL BEFORE DRILLING (findings Q6)

STATUS: ESTIMATES ONLY — DO NOT DRILL (11 values not measured yet)

VIEW FROM OUTSIDE (REAR OF THE DX) · TOP OF PANEL ↑ · PERC, CYMBAL, TOMS are further left



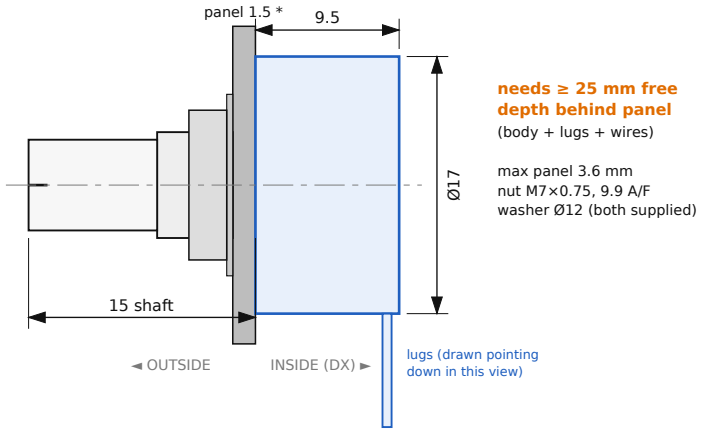
SCALE CHECK: both bars (this one and the vertical one on the left) must measure exactly 50.0 mm

HOW TO USE

1. Measure the starred (*) values (README checklist), regenerate, print at 100 %, check both bars.
2. Mains OFF and unplugged. Open the lid. Behind the SNARE pot, the blue pot body and the orange Ø24 keep-out must be clear of the tuning PCB, its screws, jack bodies and wires.
3. Cut along the dashed line and cut out the three shaft circles. Slip the template over the HI-HAT, SNARE and BASS shafts; the screw circle and label line must line up too. Tape it.
4. Cover the main board inside with paper and tape: steel swarf on the board causes shorts.
5. Centre-punch the red cross. Pilot Ø3 mm, then Ø7.5 mm drill or step drill to 8 mm (7.5–8.0 OK).
6. Tab hole only if you chose to drill it. Deburr both sides. Vacuum up every bit of swarf.
7. Fit the pot: washer + nut outside, lugs pointing as drawn. Tighten firmly, not hard.

POT BEHIND THE PANEL — side view, scale 2:1

TT Electronics / BI P160KNP-0EC15B10K — 16 mm panel pot



needs ≥ 25 mm free depth behind panel
(body + lugs + wires)

max panel 3.6 mm
nut M7×0.75, 9.9 A/F
washer Ø12 (both supplied)

VALUES USED (* = estimate — measure it, see README)

HI-HAT → SNARE shaft pitch	20.5 *
SNARE → BASS shaft pitch	20.5 *
existing shaft Ø	6.0 *
new hole below SNARE	21.0 *
new hole right of SNARE	0.0
screw right of SNARE	6.3 *
screw below SNARE	9.8 *
OUTPUTS label top below SNARE	29.9 *
panel thickness	1.5 *
behind: tuning-PCB bottom below SNARE	– *
behind: jack-body top below SNARE	– *
behind: free depth at the hole	– *
lug direction (seen from outside): right · tab hole: optional	

AUTOMATIC CHECKS (re-run after measuring)

OK	Washer (Ø12) to tuning-PCB screw head: 4.1 mm gap
OK	Washer to top of OUTPUTS label text: 2.9 mm gap
OK	Panel 1.5 mm vs. max 3.6 mm for the M7 bushing
STOP	Tuning-PCB bottom edge NOT MEASURED (pot body top at 12.5 mm)
STOP	Jack-body top NOT MEASURED (pot body bottom at 29.5 mm)
STOP	Free depth behind panel NOT MEASURED (need >= 25 mm)

LEGEND

- drill: new hole Ø7.5 (dashed: max Ø8.0)
- cut line; existing-shaft cut-outs
- pot body + lugs behind the panel
- Ø24 keep-out behind the panel (Q6)
- existing feature, use it to align
- suggested label (9 mm tape)

Dimensions in mm. Origin = SNARE shaft centre.
Seen from OUTSIDE: the BASS pot is on the right.
* = estimated from a photo. Measure, then regenerate.
Pot lugs: 1 (CCW) → J2-1 GND, 2 → J2-2 wiper,
3 (CW) → J2-3 +5 V (see wiring-diagram.pdf).
Regenerate: python3 tools/build_all.py (README.md).

PROJECT — SHEET 1 / 1 UNITS mm

DX TimeShift (snare) for Oberheim DX, board 1721A

DRAWING — A4 landscape · fits US Letter

Rear-panel drill template, SNARE DELAY pot

DWG NO.	SCALE	REV	DATE
DXTS-MECH-001	1:1 (side view 2:1)	A	2026-11-06